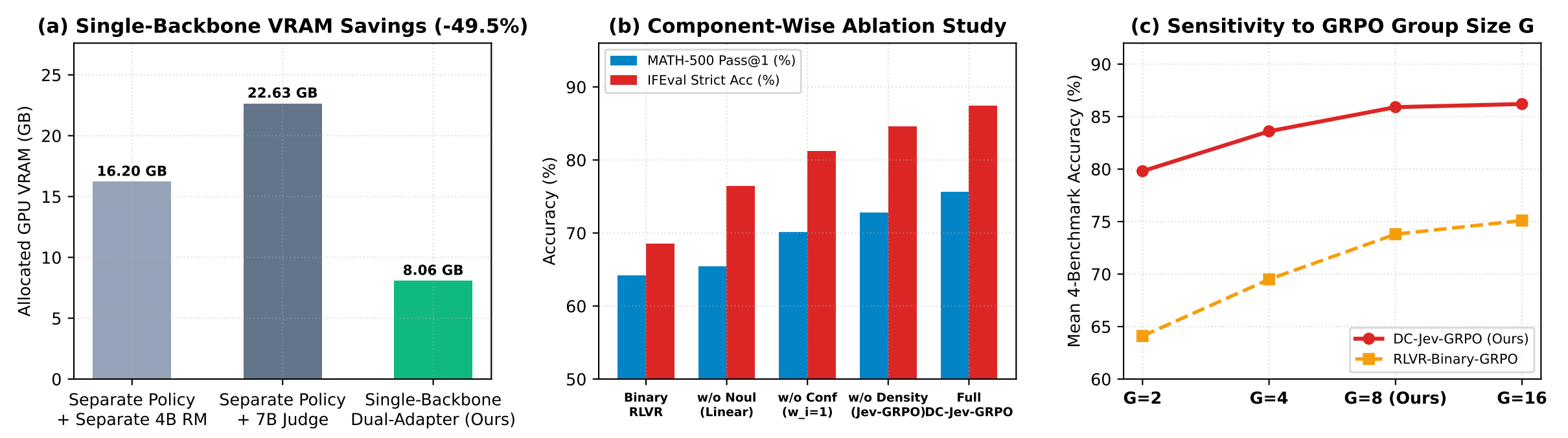
(a) Single-Backbone VRAM Savings (-49.5%)
Allocated GPU VRAM (GB)
16.20 GB
22.63 GB
8.06 GB
Separate Policy + Separate 4B RM
Separate Policy + 7B Judge
Single-Backbone Dual-Adapter (Ours)
(b) Component-Wise Ablation Study
MATH-500 Pass@1 (%)
IFEval Strict Acc (%)
Accuracy (%)
Binary RLVR
w/o Noul (Linear)
w/o Conf (w_i=1)
w/o Density (Jev-GRPO)
Full DC-Jev-GRPO
(c) Sensitivity to GRPO Group Size G
Mean 4-Benchmark Accuracy (%)
DC-Jev-GRPO (Ours)
RLVR-Binary-GRPO
G=2
G=4
G=8 (Ours)
G=16